

Sweetgum

Liquidambar styraciflua

Species background: Native to the southeastern and south-central United States, as well as parts of Mexico and Central America. Its star-shaped leaves with five to seven pointed lobes are unmistakable, and the spiky, golf-ball-sized seed pods that litter the ground beneath it in winter are just as distinctive.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

Sweetgum: Identify sweetgum's star-shaped leaf and spiky seed ball.

Standards connection: NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

Materials: Coloring sheet, crayons.

Teacher background

My leaf is shaped like a star! In fall I can show yellow, orange, red, and purple all at the same time. Watch your step under me in winter -- I drop spiky seed balls.

Procedure

- 1 Engage: Ask students to draw a star shape in the air with a finger.
- 2 Explore: Compare the leaf shape to a drawn star.
- 3 Explain: Explain that this tree's fall colors can include four different colors at once.
- 4 Art: Trace the star-shaped leaf, then color it like a nighttime sky.
- 5 Evaluate: Student can identify the leaf as star-shaped and name the spiky seed ball.

Discussion questions

What other things in nature are shaped like stars?

Assessment

Correctly identifies the star shape and the spiky fruit.

Extension

Look for a fallen seed ball on the ground beneath the tree (without touching it if spiky).

Sweetgum: Explain why sweetgum can show several fall colors on one tree.

Standards connection: NGSS 4-LS1-1; National Core Arts Standards VA:Cr2 (3-5)

Materials: Coloring sheet, colored pencils.

Teacher background

Sweetgum can show yellow, orange, red, and purple all on one tree in fall. The spiky fruit is many capsules fused together, releasing seeds through small pores.

Procedure

- 1 Engage: Ask why one tree might show more than one fall color at the same time.
- 2 Explore: Find or picture a sweetgum leaf showing more than one color.
- 3 Explain: Introduce that different pigments in the same leaf can fade at different rates.
- 4 Art: Trace a star-shaped leaf and turn it into a five-pointed star drawing -- add a face or a night sky around it.
- 5 Evaluate: Student names at least two colors this tree can show in fall.

Discussion questions

Why might a leaf have more than one color hidden inside it all along?

Assessment

Names two or more correct fall colors.

Extension

Collect leaves in as many different colors as you can find from this tree.

Sweetgum: Explain independent pigment breakdown as the cause of sweetgum's multicolor fall display.

Standards connection: NGSS MS-LS1-5; National Core Arts Standards VA:Cr2/Re7 (6-8)

Materials: Coloring sheet, paint in yellow, red, and purple.

Teacher background

Chlorophyll, carotenoids, and anthocyanins fade at different rates, producing multi-color canopies. The spiky fruit is many capsules fused together, releasing seeds through small pores.

Procedure

- 1 Engage: If a leaf has three different pigments, why wouldn't they all disappear at the same time?
- 2 Explore: Research the three main leaf pigments and their typical breakdown order.
- 3 Explain: Connect independent pigment breakdown rates to sweetgum's dramatic multicolor display.
- 4 Art: Create a fall-color study of the same star-shaped leaf outline, painted three different ways to show yellow, red, and purple variants -- research why one tree can show all three.
- 5 Evaluate: Write a paragraph naming the three pigments and explaining why they don't fade in sync.

Discussion questions

Why might natural selection not have 'streamlined' all trees to lose their pigments at exactly the same rate?

Assessment

Paragraph correctly names all three pigments and explains the mismatched fade rates.

Extension

Track one sweetgum leaf's color change over several weeks if possible.

GRADE BAND 9-12

Sweetgum: Design an observational study tracking pigment change and hypothesize which pathway dominates first.

Standards connection: NGSS HS-LS1-5; National Core Arts Standards Proficient-level Creating/Responding (9-12)

Materials: Coloring sheet, notebook or spreadsheet for weekly logging, dissecting tools for the seed capsule.

Teacher background

Chlorophyll, carotenoids, and anthocyanins break down or synthesize at different rates and are influenced by different environmental triggers -- temperature, sunlight, and sugar accumulation. Design a simple observational log tracking this tree's color change over several weeks and hypothesize which pigment pathway dominates first.

Procedure

- 1 Engage: What environmental variables would you need to track alongside leaf color to test a hypothesis about pigment timing?
- 2 Explore: Design a weekly observation log format for tracking this tree's color change.
- 3 Explain: Research which environmental triggers (temperature, sunlight, sugar) affect each pigment pathway.
- 4 Art: Do a botanical cross-section sketch of the spiky seed capsule (many fused capsules), labeling the small pores seeds are released through, following scientific illustration conventions.
- 5 Evaluate: Submit the observation log design, a stated hypothesis, and the labeled seed-capsule sketch.

Discussion questions

How would you distinguish a temperature-driven pigment change from a daylight-length-driven one in your data?

Assessment

Rubric covering log design rigor, hypothesis clarity, and illustration accuracy.

Extension

Compare sweetgum's multi-pigment fall display to red maple's more uniform, actively-produced red color.

SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.